Week 2

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1 Prompt Engineering

1.1 Type of prompts

- **Direct Prompts**: clear and specific instructions; best for simple tasks when you know exactly what output you need the AI to generate.
- **Open-ended Prompts**: diverse outputs without any constraints; for brainstorming, storytelling, or exploratory discussions.
- **Task-specific Prompts**: precise, goal-oriented tasks; defined focus and have additional context or examples to help generate responses.

1.2 Prompt Engineering Techniques

- **Zero-shot prompting**: response based on pre-trained knowledge without any examples or hints in the prompt; for simple, familiar, and straightforward tasks.
  - They can be less effective for tasks with domain knowledge, high understanding and specific outputs; they can have biases in training that can affect fairness and accuracy.
- **One-shot prompting**: model is given a single example along with a task prompt to guide its response; for cases where collecting large amounts of data is impractical.
  - Add sample input and output, and the instruction in the prompt.

  - They can have biases because one example shape the output, and not work for complex tasks or tasks that need domain knowledge.

- **Few-shot prompting**: model is given a few examples along with a task prompt to guide its performance; helps to find patterns and improve accuracy of response.
  - Add sample inputs and outputs, and the instruction in the prompt.
  - They need to designed carefully and require domain knowledge to create effective examples so that the model can pick up the right things; more intensive as more examples increase processing time.

- **Chain-of-thought prompting**: technique to enhance the output of LLM by guiding a model through a step-by-step reasoning process by using a coherent series of logical steps; useful for when prompting the model to perform tasks with logic or maths.
  - Use a checklist and include these in the prompt: i) a clear task or problem statement, ii) a reasoning trigger or phrase that cues AI to reason, iii) a step-by-step format to organize the reasoning, iv) a format to summarize the final answer.
  - They are less effective for tasks with complex or branching logic as the model follows a liner reasoning chain; best with instructions are clear and guide effectively, as vague instructions = poor output.

- **Meta prompting**: technique where model is asked to generate or improve its own prompt; to help enhance the quality, focus or consistency of output so that it will be more accurate, polished with the required tone, style or level of detail.
  - Add instruction and guidance for refinement to the prompt.
  - They need well-structured guidance to refine them; add complexity and processing time as the model should analyze and modify a prompt before generating a response.

1.3 Mistakes to avoid when prompting

- Vague Prompts

- Grammatical errors
- Leading questions
- Context gaps
- Complex Prompts

1.4 Best Practices

- Start with a clear objective
- Provide context
- Be specific and precise
- Use a conversational tone
- Refine and iterate

2 Chatbots

2.1 Types of Chatbots

- **Text-based**: queries only through text-based interactions; help in customer service handling various routine queries like order status, and detail updates.
- **Voice-based**: spoken commands using speech recognition to interpret spoken query and responds verbally; can handle complex interactions and adapt to various accents and speech patterns.
- **Virtual assistants**: provide a smooth and multifunctional UX; for a variety of tasks like answering emails, tracking orders, scheduling appointments, weather updates and controlling smart home devices.

2.2 Conversational AI

- It enables machines to engage in human-like conversations with users.
- Uses NLP, machine learning, and sentiment analysis.
  - **Natural language Processing**: understand and interpret human language; to process the words and phrases you use, understand the meaning, and determine the context.
  - **Machine learning**: learn from data and improve its performance over time by learning from past interactions, recognize patterns, and improve responses over time.
  - **Sentiment analysis**: detect and understand the emotional tone; mood detector and enables to identify whether the message is positive, negative, or neutral.
- This allows the chatbot to perform tasks like interpreting human language, remembering past conversations, incorporating context into their responses, and creating more intuitive and effective interactions.

2.3 Designing Chatbots

- **User Interface**: designing the chatbot personality, and engagement
  - **Chatbot personality**: giving it unique characteristics and communication style that aligns with the company identity; good one helps users connect with the chatbot and improves their experience.
    - **Appearance**: layout such as colours, buttons, and fonts.
    - **Interaction**: user experience design; focusing on the user's interaction with the chatbot, the information that they have, and the information that they recieve.
    - **Dialog**: responses for the chatbot; writing clear, helpful, and engaging text and matching the company brand with the tone and personality.
    - **Learning**: the ability to adapt to different user actions; involves machine learning which helps the chatbot to interpret accurately and give personalized responses; with NLP and

ML, you train the chatbot to interpret users' question and answers and respond.

  - **Engagement**: making the chatbot interactive and easy to use so that uses want to interact with it; easy to use, responsive, interactive, and reflection of the company brand.

- **Meeting customer needs**

  - **Define the goal**: what you want it to do; keeps the focus on a specific need
  - **Choose the starting domain**: specific area to focus on initially; eg billing section of a website, so domain to integrate the chatbot to effectively.
  - **Choose the starting channel**: platform that is the most frequently visited by customers.
  - **Start small**: assigning a small, manageable task to the bot to test its functionality; how well it performs and gather feedback.
  - **Identify key topics**: focus on few most-asked specific questions or tasks that it will handle to ensure that it give the most accurate information for them.
  - **Identify relevant content sources**: take stock of all the helpful content such as product information, articles, and FAQs within the domain, and set up a quick search function.
  - **Plan for handoff**: when chatbot can not handle a request, plan to escalate to a human agent.

- **Design flow logic**: how it reacts to each statement made during an exchange and proceeds to the next step; flow logic needs simple if-else statements, decision trees, complex algorithms, or machine learning-driven probabilistic logic.